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PI Name:	Mansur, Louis K. Ph.D.		
Project Title:	Materials Science and Particle Transport Assessment of Novel and Multifunctional Shielding Materials		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	37831-6138	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Assignment:	NOTE: Received NCE to 01/31/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
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Task Description:	Our approach consists of 1.) Specially preparing advanced shielding materials, 2.) Characterizing their structure and properties, 3.) Carrying out GCR transport and beam transport calculations for both pre-experiment planning and post-measurement analyses, and 4.) Measuring particle types and energy spectra during beam irradiation experiments at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory. In-depth assessments will be carried out with respect to physical, mechanical and shielding properties. Properties such as structural strength, hydrogen storage, potential for thermal management, and electrical attributes will be assessed on a material specific basis. Calculations of shielding effectiveness will be accompanied by high energy accelerator experiments at the NASA Space Radiation Laboratory and other high energy heavy-ion accelerators. A primary aim of these experiments is to determine shielding effectiveness. An equally important goal is to measure the response of the materials to environmental variables. For example, thermal transport and mechanical properties will be measured. There is a high priority need to evaluate new materials for spacecraft shielding in deep space missions. This research will carry out new work on space radiation shielding materials. Both novel shielding materials and multifunctional spacecraft materials that incorporate desirable shielding properties will be investigated. The aim is to clearly establish the most promising materials and to assess their key properties with respect to space radiation shielding and suitability for other spacecraft applications. The approach will emphasize materials science. The required nuclear properties, in particular the overriding need to incorporate the highest possible proportion of light nuclei, especially hydrogen, will be kept in constant view. The identification, development and qualification of new and multifunctional shielding materials are of the high importance to future manned NASA space missions. Knowledgeable researchers understand that new shielding materials must be developed. Without better shielding for spacecraft crew, long term deep space missions would not be possible. Based on new understanding it is now clear that only those shielding materials that contain high proportions of hydrogen and other light elements, and with an absolute minimum of heavy elements, can be effective against galactic cosmic rays. It is a disturbing realization that recent transport calculations for galactic cosmic rays (GCR), coupled with currently proposed calculational models of biological effects have led to an understanding that conventional spacecraft materials like aluminum alloys may result in radiation effects that are more detrimental than would result with no shielding at all. This effect occurs through fragmentation of the incoming cosmic rays and the heavy nuclei of the spacecraft material such that the sum of the biological damage from the resulting new charged particles, neutrons and energetic photons exceeds that of the original cosmic ray particle. Conventional models of biological effect are based on risk assessment dictated by the magnitude of linear energy transfer (LET) of a particular particle at a given energy, coupled with accepted quality factors that calibrate LET in terms of relative biological effectiveness (RBE) in relation to X-rays (ICRP, 1991). It can be characterized as a dose equivalent method. While this approach may often be useful as a first approximation, it cannot recognize particular mechanisms of action of radiation on organisms. Heavy charged particles may interact with cellular molecules and subassemblies in many different ways, producing detrimental effects that are physically inaccessible to X-rays no matter how high the X-ray dose. An example based on radiation damage physics even without considering a detailed biological pathway is that charged particles with energies as low as a few keV may displace atoms from their molecular sites or cellular subassembly sites to locations far removed, in addition to producing ionization and excitation. X-rays on the other hand cannot generally displace atoms. They produce ionization and excitation. [For a photon to produce displacements to a measurable degree, it generally must be in the energy range of gamma rays above 2 MeV (Mansur and Farrell, 1997). While the gamma ray does not produce displacements directly, the resulting Compton electrons and electron/positron pairs can have enough energy to displace atoms.] Thus, there are compelling physical reasons on the most basic level that heavy charged particles may have different and more severe effects than would be obtained by any multiple of X-ray dose.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Exploration in deep space and missions to Mars are expected to have major benefits on earth. By better understanding the solar system radiation environment and by developing new technologies to protect and support life in hostile conditions, the quality of life on earth with respect to radiation protection can be improved. New materials are necessary to shield spacecraft crew from potentially lethal doses of radiation from galactic cosmic rays. In turn these materials will 1.) bring the benefits to earth that are the outcome of successful missions to space, and 2.) provide an important new suite of materials with novel and multifunctional properties to engineers and designers for earthbound applications.
Task Progress:	During 2004 and 2005 we have been carrying out investigations of properties of two classes of materials. These are carbon forms infiltrated with hydrocarbons, and polymer fibers and fiber-resin composites. Both classes of materials have the potential to serve one or more functions in spacecraft in addition to GCR shielding. Brief descriptions of the materials are given below. Expected reductions in dose and dose equivalent have been calculated for a wide range of thickness of these materials. Initial shielding experiments have been carried out using high energy heavy ion beams at NSRL. Experiments were carried out to infiltrate carbon materials with polyethylene (PE) and to measure the level of methane adsorption as a function of temperature. Both PE and methane contain high proportions of hydrogen to carbon, 2:1 and 4:1 respectively. The PE-infiltrated forms were utilized in subsequent beam irradiation experiments. A set of polymer fibers and fiber-resin composites were also prepared. In the case of the carbon forms, the experimental materials were fabricated by our research group. For the polymer fibers and composites, many of the experimental materials were already available in the desired forms. Specimens each material were prepared as multiple disks or layers of fabric, which allowed the thickness exposed to beam to be varied according to the materials irradiation plan. Three classes of carbon materials were explored: carbon foams; carbon fiber monoliths; carbon powder. The foams and monoliths were infiltrated with a high density polyethylene (HDPE). Scanning electron microscopy of these materials was carried out before and after infiltration. Polymeric materials of several forms were utilized in this work. These included fabrics of Kevlar, Spectra, and Zylon, and resin composites of two of these fibers denoted as Spectra epoxy, Spectra Shield, and Zylon/PVB phenolic. In addition, a material denoted as IM 7/epoxy, which is a carbon fiber/resin composite was included. Specimens of LDPE and of an epoxy were also included as reference materials. All the materials described were irradiated at the NSRL with a 1 GeV/nucleon O16 beam. In addition, the Kevlar fabric, PE-infiltrated low density carbon foam, and PE-infiltrated carbon fiber monolith were irradiated with an O16 beam at

	energy of 0.6 GeV/nucleon to serve as tie markers to a significantly different beam condition. In general the materials were irradiated at a standard areal density of approximately 5 g/cm2. Three of the materials were also irradiated at approximately one-third of this thickness. By separating or adding together specimen stacks, a thickness study was carried out on the PE-infiltrated carbons that ranged from approximately 1.8 to 15 g/cm2 using the higher energy beam. From the point of view of nuclear interactions with such high energy beams, the structure and bonding in these materials is irrelevant. It is the bulk averaged nuclear composition that determines their shielding characteristics. Important metrics are the proportion of the non-interacting primary beam and the fragmentation of the primary beam ions by interactions with nuclei in the shield, as measured by the flux and energy distribution of ions that emerge from the specimen stack. These in turn are used to determine the dose and dose equivalent that would have been received by organisms behind the shielding material. HZE transport calculations have been completed or are in progress prior to the completion of the analysis of the measurements to predict the performance of the materials with respect to shielding metrics including dose (completed), dose equivalent (completed), transmitted primary beam, and beam fragment fluence spectra (in progress). Measurements of thermal transport and mechanical properties of the carbon foams before and after infiltration with HDPE have been carried out.
Bibliography Type:	Description: (Last Updated: 08/08/2007)
Abstracts for Journals and Proceedings	Mansur, L. K., Frame, B. J., Gallego, N. C., Guetersloh, S. B., Johnson, J. O., Klett, J. W., Townsend, L. W. "High Energy O16-Beam Shielding Performance of Carbon and Polymer Materials" 16th Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York Workshop Abstract Book , May-2005
Abstracts for Journals and Proceedings	Mansur, L. K., Burchell, T.D, Frame, B. J., Gallego, N. C., Guetersloh, C. J. Janke, J. W., Johnson, J. O., Townsend, L. W. "Materials Science and Particle Transport Assessment of Novel and Multifunctional Shielding Materials" Third International Workshop on Space Radiation Research, Port Jefferson, New York Workshop Abstract Book , May-2004
Articles in Peer-reviewed Journals	Mansur, L. K., Frame, B. J., Gallego, N. C., Guetersloh, S. B., Klett, J. W., Johnson, J. O., Townsend, L. W. "Assessment of Shielding Material Performance for Deep Space Missions" Proc. Materials Research Society, Symposium NN, Materials for Space Applications, Boston, MA, Nov. 29 – Dec. 3, 2004, volume 851, pp. 375-380, Materials Research Society, Warrendale, PA. , Mar-2005
Presentation	Mansur, L. K., Burchell, T.D, Frame, B. J., Gallego, N. C., Guetersloh, C. J. Janke, J. W., Johnson, J. O., Townsend, L. W. "Progress in Materials Science and Particle Transport Assessment of Novel and Multifunctional GCR Shielding Materials" Space Radiation Shielding Principal Investigators Meeting National Space Science and Technology Center Huntsville, AL Aug-2004
Presentation	Mansur, L. K., Frame, B. J., Gallego, N. C., Klett, J. W., Townsend, L. W. "Progress in Materials Science and Particle Transport Assessment of Novel and Multifunctional GCR Shielding Materials" Marshall Space Flight Center Huntsville, AL Oct-2004
Presentation	Mansur, L. K. "Assessing Performance of GCR Shielding Materials for Deep Space Missions" Seminar at the National Space Science and Technology Center, Huntsville, AL Jul-2005